

SurgiRAG: A Domain-Adaptive Retrieval-Augmented Generation Framework for Procedural Text Understanding in Surgery

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Abstract

Surgical education and intra-operative support require systems capable of grounding language understanding in temporally aligned, procedurally rich multimodal data. While Retrieval-Augmented Generation (RAG) has shown promise in textual domains, its extension to high-fidelity, procedure-specific content remains underexplored. In this work, we introduce SurgiRAG, a retrieval-augmented generation framework designed for laparoscopic cholecystectomy procedures. SurgiRAG integrates surgical transcripts and domain-specific documentation to support question answering and procedural explanation. The architecture employs a retrieval system using BioBERT-based embeddings for both video transcript and textual context processing. Given a natural language query e.g., “How is the cystic duct isolated?” - SurgiRAG retrieves semantically aligned segments to generate grounded responses. By leveraging the complementary nature of real-time surgical narration and structured documentation, SurgiRAG captures both procedural nuance and clinical precision. Preliminary results demonstrate substantial gains in factual accuracy and temporal relevance over baseline models, highlighting the promise of SurgiRAG for surgical training, intra-operative guidance, and post-operative documentation.

1 Introduction

The rise of foundational models has fundamentally reshaped the landscape of artificial intelligence. Large Language Models (LLMs), such as GPT-4 (OpenAI et al., 2023), Gemini (Google

Deepmind et al., 2023), and LLaMA (Touvron et al., 2023), have demonstrated remarkable capabilities across a wide range of natural language processing (NLP) tasks including question answering, logical reasoning, multilingual translation, in-context learning. However, despite their impressive generalization abilities, LLMs suffer from critical limitations: their knowledge is frozen at training time, their reasoning may lack grounding in verifiable evidence, and they remain prone to hallucinations, especially in high-stakes, domain-specific applications such as medicine.

To address some of these shortcomings, Retrieval-Augmented Generation (RAG) has emerged as a promising framework that combines non-parametric document retrieval with generative modeling, enabling LLMs to ground their outputs in external, verifiable sources (Lewis et al., 2020). Beyond Text (Riedler et al., 2024) explores how RAG can be optimized for industrial domains by incorporating visual documentation alongside textual manuals, showing improvements in retrieval relevance and factual grounding.

Despite this progress, high-stakes, real-world procedural domains such as surgery remain underexplored. Existing multimodal RAG systems primarily focus on static visual artifacts (e.g., diagrams, screenshots) or video understanding in general-purpose contexts. In contrast, surgical question answering requires tight temporal grounding, domain-specific terminology, and the ability to reason over fine-grained procedural knowledge embedded in long-form video and technical documentation. Surgical video-language understanding is a uniquely valuable and underdeveloped testbed for multimodal retrieval-augmented generation. Current surgical AI systems either operate purely on visual signals (e.g., phase recognition) or rely on static documentation and pre-scripted rules - failing to dynamically integrate procedural knowledge with video context

in real time.

In this work, we introduce SurgiRAG, a retrieval-augmented generation framework tailored to the surgical domain. SurgiRAG is designed to answer natural language questions by leveraging two complementary information sources: (1) video transcripts containing real-time procedural narration from surgical videos, and (2) domain-specific textual documentation such as operative guidelines and surgical protocols. This dual-text modality enables the system to meet the unique demands of surgical question answering: temporal grounding, procedural reasoning, and clinically precise language use, without requiring direct visual processing. At inference time, the top-ranked segments from both text modalities are retrieved using FAISS and fused to construct a coherent, grounded context. A transformer-based generator (LLaMA) then synthesizes the final response.

To our knowledge, SurgiRAG is the first RAG system designed specifically for procedural surgical AI. We benchmark our framework on laparoscopic cholecystectomy procedures and demonstrate that it significantly outperforms baseline QA models across metrics of factual accuracy, temporal relevance, and retrieval precision. Our findings represent a concrete step toward clinically deployable, context-aware AI assistants in surgical and other high-stakes domains.

2 Methods

3 Methods

Our system follows a modular Retrieval-Augmented Generation (RAG) architecture tailored to the surgical QA domain. The pipeline consists of three main components: (1) a retriever trained to identify semantically relevant text chunks from a large corpus of surgical documents, (2) a cross-encoder reranker that reorders the top retrieved results for precision, and (3) a generator fine-tuned to produce grounded and coherent answers from context-aware prompts. In this section, we detail the datasets used for training and evaluation, our preprocessing pipeline for structured document ingestion, and the retrieval and generation methods developed for this task.

3.1 Datasets

We used a two-stage data split for model development and evaluation. For training and proto-

typing, we relied exclusively on a comprehensive surgical guide titled "How do I do it: laparoscopic cholecystectomy" (Majumder et al., 2020), published in the *Annals of Laparoscopic and Endoscopic Surgery*. This expert-written article offers detailed coverage of procedural steps, anatomical considerations, and safety protocols, making it an ideal resource for simulating high-quality surgical question answering. To further enhance domain grounding and biomedical QA performance, we fine-tuned our retriever and generator components on the BioASQ QA dataset, which provides expert-annotated biomedical question-answer pairs across various medical subdomains.

For evaluation, we curated a held-out test set comprising 17 documents, including processed transcripts from four laparoscopic cholecystectomy videos (spanning instructional walkthroughs, clinical cases, and educational presentations) and formal surgical documentation from authoritative sources. This test set totaled 409 processed pages segmented into 1,527 text chunks. Importantly, there was no overlap between training and testing materials.

We construct a custom 32-item QA benchmark covering procedural steps, anatomical landmarks, safety measures, and intraoperative decisions. Questions were generated using GPT to ensure consistency in linguistic style and topic coverage. Importantly, the benchmark was created with no content overlap with training materials, maintaining a clear separation between pretraining and evaluation data.

3.2 Data Preprocessing

For the video processing, we employed AssemblyAI’s speech-to-text API to extract high-quality transcripts from surgical videos, achieving confidence scores between 0.95-0.98 across all content. Transcripts were automatically segmented based on natural speech patterns and procedural phases, with each segment containing precise timestamp information for temporal grounding. These processed transcripts were then converted to PDF format for integration with the document corpus. All original surgical PDFs and converted transcript PDFs were processed using a unified pipeline that extracts text content and converts it into structured page-level representations. Each processed page contains metadata including document source, page number, character count, word

count, and sentence-level segmentation for fine-grained analysis.

Documents were segmented using a token-aware chunking pipeline that creates coherent 300-token segments with appropriate overlap to maintain contextual coherence while ensuring comprehensive coverage. Text cleaning included normalization of medical terminology and removal of formatting artifacts to optimize retrieval performance.

All resulting text chunks were embedded using a fine-tuned BioBERT-based SentenceTransformer, which was previously adapted on biomedical QA data from BioASQ. This model generated 768-dimensional embeddings optimized for medical semantic similarity. The final collection of 1,527 chunk embeddings was indexed using FAISS with an inner product search strategy (IndexFlatIP), allowing for fast and efficient similarity-based retrieval at inference time.

3.3 Model Architecture

SurgiRAG is composed of three tightly integrated components: a domain-adapted dense retriever, a cross-encoder reranker, and a fine-tuned generator. Each module is optimized for the surgical QA setting, enabling faithful, context-grounded response generation over heterogeneous sources including procedural guides and surgical video transcripts.

For retrieval, we employ a fine-tuned BioBERT-based SentenceTransformer, initialized from the pritamdeka/BioBERT-mnli-snli-scinli-scitail-mednli-stsb model. This version of BioBERT was selected for its specialized understanding of biomedical terminology and its pre-training on PubMed abstracts and PMC full-text articles. This retriever was adapted on domain-specific question-document pairs using BioASQ data to better capture the semantic structure of medical and procedural queries. Both the surgical corpus and incoming queries are encoded into a shared 768-dimensional semantic space. To ensure robust semantic alignment, the same encoder is used for both corpus chunking and query processing. We index all chunk embeddings using FAISS (IndexFlatIP) to enable efficient inner product-based retrieval at inference time.

To improve the relevance and precision of retrieved evidence, we implement a cross-encoder reranker using the BAAI/bge-reranker-large model. The reranker processes the top-

k retrieved passages alongside the query and computes refined relevance scores using cross-attention mechanisms. This module is essential for filtering semantically similar but clinically irrelevant passages and for prioritizing content that precisely answers the question. All final context inputs to the generator are drawn from the top reranked results, making this component a critical part of the system’s factual grounding strategy.

For generation, we use LLaMA-3.2-11B-Vision-Instruct as our backbone model, configured in a text-only mode. Although this model supports multimodal inputs, we restrict our implementation to text-only inputs derived from video transcripts and surgical documents. This design choice was deliberate: the architecture’s native support for vision-language modeling positions SurgiRAG for seamless expansion into future multimodal retrieval-augmented generation, where visual features such as surgical frames, instrument views, or anatomical diagrams can be integrated into the pipeline. To adapt the generator for domain-specific biomedical QA, we fine-tuned it using LoRA (Low-Rank Adaptation) on the same curated question-answer pairs from the BioASQ dataset as the retriever fine-tuning process. This fine-tuning improves the model’s ability to interpret and synthesize grounded answers from retrieved medical context. The LoRA adapters were trained using the PEFT framework and remain unmerged during inference to avoid the decoding instability we observed during early weight-merging experiments. All generation is performed using bfloat16 precision, balancing computational efficiency with high-quality text generation.

3.4 Experimental Design

To evaluate the contribution of each system component to overall performance, we conducted a systematic ablation study that varied the retrieval backbone, reranking strategy, and generation model. This analysis allowed us to isolate the effect of each stage and assess its impact on factual accuracy, semantic grounding, and fluency.

We explored a $3 \times 2 \times 2$ full-factorial design, resulting in 12 distinct pipeline configurations. Each configuration was evaluated on the same set of surgical QA prompts using consistent random seeds and few-shot prompting to ensure comparability across variants.

- **Random baseline:** Passages are selected

uniformly at random from the corpus. This serves as a naive lower bound to contrast against learned retrievers.

- **Base BioBERT:** A pre-trained BioBERT SentenceTransformer without domain-specific fine-tuning, serving as a general biomedical retriever.
- **Fine-tuned BioBERT:** Our domain-adapted retriever, fine-tuned on question–document pairs from the BioASQ dataset to improve semantic alignment for surgical queries.

3.4.1 Reranking Variants

- **No reranking:** Retrieved passages are passed to the generator based solely on dense similarity scores from the retriever.
- **Cross-encoder reranking:** A BGE large cross-encoder model re-ranks the top- k retrieved passages by directly comparing each with the query using cross-attention. This reranking step is included in our final system and consistently improves contextual precision and factual grounding.

3.4.2 Generation Variants

- **Base LLaMA:** The unmodified LLaMA-3.2-11B-Vision-Instruct model used in a text-only setting without biomedical adaptation.
- **Fine-tuned LLaMA:** A LoRA-adapted version of LLaMA, fine-tuned on BioASQ QA data to better synthesize grounded answers in medical contexts. This version is used in our best-performing system.

All combinations were evaluated under identical conditions to measure their relative contributions to performance metrics, including BERTScore, ROUGE-L, BLEU, GPT-4o-mini judgment scores, and faithfulness. This ablation framework forms the foundation for our subsequent results and analysis.

3.4.3 Evaluation Methodology

We implemented a multi-tiered evaluation framework to assess the quality, faithfulness, and clinical relevance of generated responses across all system variants. Our evaluation combined standard NLP metrics, factuality benchmarks, and learned model-based judgments to provide a comprehensive view of system performance. For standard overlap-based metrics, we used:

- **BLEU:** Measures n-gram overlap between generated and reference answers, capturing surface-level similarity.
- **ROUGE-L:** Computes the longest common subsequence F1-score, reflecting content preservation.
- **BERTScore:** Assesses semantic similarity using contextualized embeddings to evaluate meaning alignment between generated and reference responses.

To assess factual consistency between retrieved evidence and generated outputs, we applied two complementary methods:

- **FactCC:** A model trained to detect factual inconsistency between input context and output summaries.
- **BART-large-MNLI:** Used to evaluate entailment between the retrieved context and generated answers, with entailment probability serving as a soft proxy for groundedness.

Recognizing the limitations of static metrics in specialized domains like surgery, we further incorporated GPT-4o-mini as an automated evaluation judge. Each generated response was assessed along four axes:

- **Context Precision:** The degree to which the retrieved passages are relevant to the input query.
- **Faithfulness:** How accurately the generated answer reflects the retrieved context without hallucination.
- **Fluency:** Grammatical correctness and natural language flow, scored on a 1–5 scale.
- **Coherence:** Logical structure and reasoning quality, also scored on a 1–5 scale.

3.5 Results and Discussion

Our ablation study across 12 system configurations revealed clear patterns in how different components affect performance. The results show that semantic retrieval is essential for faithfulness, but comes with trade-offs in text fluency that have important implications for medical AI systems.

Retriever	Reranker	Generator	BLEU	ROUGE-L _{F1}	BERTScore _{F1}	FactCC _{acc}	FACT _{conf}	Entailment _{avg}	GPT-Prec@5	GPT-Faith	GPT-Fluency	GPT-Coherence
Random	None	Base	0.0058	0.0747	0.7764	0.3750	0.3874	0.2858	0.0000	0.2500	3.7500	4.1562
Random	None	Finetuned	0.0068	0.0605	0.8016	0.6250	0.6439	0.8649	0.0000	0.0938	3.0938	2.8438
Random	Cross-Encoder	Base	0.0063	0.0612	0.8029	0.7500	0.7090	0.8315	0.0000	0.1250	3.1250	2.7500
Random	Cross-Encoder	Finetuned	0.0065	0.0603	0.8030	0.7188	0.7037	0.8703	0.0000	0.0625	3.0000	2.7188
Base	None	Base	0.0091	0.0701	0.8095	0.5625	0.5918	0.7825	0.0938	0.5938	3.1875	3.2500
Base	None	Finetuned	0.0082	0.0679	0.8059	0.7188	0.6379	0.8206	0.0938	0.5625	3.2188	3.0625
Base	Cross-Encoder	Base	0.0071	0.0716	0.8089	0.5938	0.5749	0.7935	0.2500	0.5625	3.1250	3.0625
Base	Cross-Encoder	Finetuned	0.0122	0.0731	0.8112	0.6250	0.5907	0.7455	0.2812	0.6250	3.1875	3.0938
Finetuned	None	Base	0.0060	0.0660	0.8073	0.6250	0.5980	0.7872	0.0625	0.3750	3.1875	3.1250
Finetuned	None	Finetuned	0.0063	0.0685	0.8066	0.6250	0.5806	0.7747	0.0625	0.4375	3.3125	3.1875
Finetuned	Cross-Encoder	Base	0.0082	0.0676	0.8101	0.4375	0.4538	0.7121	0.1250	0.6250	3.3438	3.2500
Finetuned	Cross-Encoder	Finetuned	0.0077	0.0683	0.8121	0.4375	0.4621	0.7334	0.1250	0.6562	3.0938	3.1875

Table 1: Ablation study across retriever, reranker, and generator configurations.

The top-performing system was the full SurgiRAG pipeline, consisting of a fine-tuned retriever, BGE cross-encoder reranker, and LoRA-fine-tuned LLaMA generator. This configuration achieved the highest semantic similarity with a BERTScore F1 of 0.8121, and also outperformed all others in faithfulness, as judged by GPT-4o-mini (0.6562). This indicates that domain adaptation of both the retriever and generator, combined with high-precision reranking, leads to the most contextually accurate and well-grounded answers.

In contrast, configurations using random retrieval consistently underperformed on grounding-sensitive metrics, with faithfulness scores ranging from 0.0625 to 0.2500. Interestingly, several of these systems still achieved BERTScores above 0.80, comparable to those of the top semantic configurations, despite lacking meaningful retrieval. This discrepancy reflects a known limitation of surface-level similarity metrics: they can overestimate answer quality when the generator relies on internal pretrained knowledge rather than retrieved evidence. These inflated BERTScores are likely influenced by the strength of our generator, LLaMA-3.2-11B-Vision-Instruct, which represents a state-of-the-art open-source foundation model, as well as our high-performing BioBERT-based retriever, which may contribute domain-specific fluency even in less informative retrieval scenarios. For example, the configuration using random retrieval + cross-encoder reranker + fine-tuned generator achieved a BERTScore of 0.8030, but a faithfulness score of only 0.0625, illustrating the model’s ability to produce fluent but ungrounded responses in the absence of relevant context.

The gap in GPT-assessed faithfulness between the SurgiRAG model (0.6562) and the best-performing random variant (0.2500) spans more than 40 percentage points, underscoring the critical role of effective retrieval and reranking for clinical and procedural QA. All systems with se-

mantic retrieval and reranking significantly outperformed their random counterparts, confirming that high-quality evidence selection is essential for generating medically reliable responses.

These findings also illustrate the limitations of conventional metrics like BLEU and ROUGE, which showed minimal variance across configurations. In contrast, LLM-based evaluation surfaced meaningful differences in factuality, coherence, and faithfulness, making it a more suitable proxy for human judgment in domain-specific tasks. Overall, these results affirm that retrieval quality and context-grounded generation are far more important than stylistic fluency in medical question answering systems.

3.5.1 Component Impact Analysis

Breaking down each component’s contribution, we found that retriever choice had the biggest impact on system performance. Any semantic retriever (base or fine-tuned BioBERT) dramatically outperformed random selection, improving faithfulness by 25-40 percentage points. However, the difference between base BioBERT and our fine-tuned version was modest, suggesting that BioBERT’s extensive biomedical pre-training provides most of the benefit. Cross-encoder reranking consistently improved performance, adding 5-15 percentage points to faithfulness across most configurations. The BGE reranker’s ability to directly compare query-document pairs provides a significant advantage over simple embedding similarity, though it does increase computational overhead substantially. Generator fine-tuning proved essential for faithful context utilization. Our LoRA-adapted LLaMA model outperformed the base version by 5-20 percentage points on faithfulness, demonstrating that domain-specific training helps the model better ground its responses in retrieved evidence.

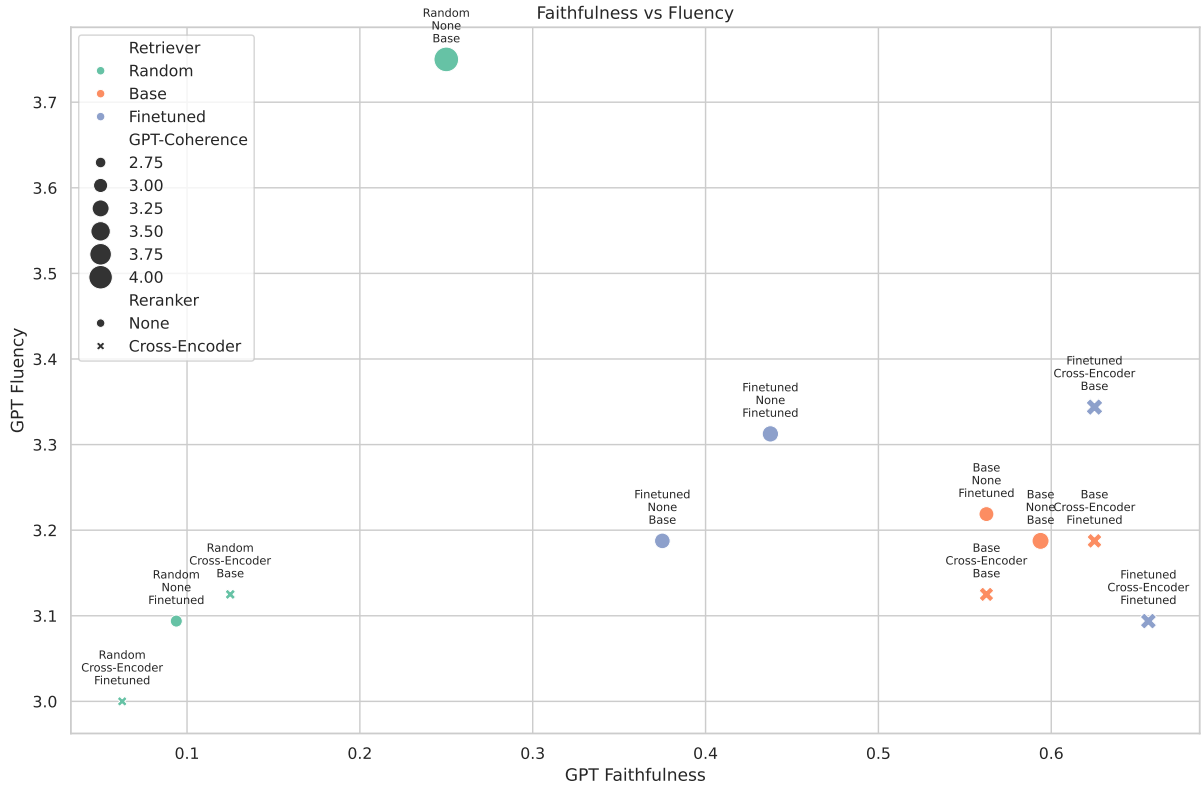


Figure 1: Faithfulness vs. Fluency trade-off across all 12 configurations.

3.5.2 Fluency vs. Faithfulness Trade-off

We uncovered a striking fluency-versus-faithfulness trade-off (Fig.1). The Random Retriever + None + Base Generator baseline - whose retrieved evidence is essentially noise, achieved the highest GPT-assessed fluency (3.75) and coherence (4.16) of any system. By contrast, our best grounded model (Finetuned Retriever + Cross-Encoder + Finetuned Generator) delivered far superior faithfulness (0.66) but a lower fluency score of 3.09 and coherence of 3.19.

When retrieval is random, the generator largely ignores the irrelevant context and falls back on its well-trained internal priors, producing smooth but hallucinated prose. When meaningful passages are retrieved, the generator actively weaves those snippets into the answer; this grounding step occasionally introduces lexical repetition or choppy transitions, reducing surface-level polish.

This finding has serious implications for medical AI applications. For surgical education and decision-support scenarios, this result underscores a critical design principle: accuracy must outweigh eloquence. A slightly less fluent answer that is verifiably grounded in the operative protocol is vastly preferable to a perfectly-worded but

clinically-wrong response.

3.5.3 Secondary Metrics and Validation

Our FactCC accuracy results (0.37 - 0.75 range) were less discriminative than expected, with some random systems matching semantic ones. This suggests that FactCC, trained on general text, may not capture domain-specific factual requirements that our GPT-4o-mini judge identifies. The closer alignment between GPT faithfulness scores and expected system performance indicates that LLM-based evaluation may be more appropriate for specialized domains. The coherence scores (2.72 - 4.16 range) followed similar patterns to fluency but with smaller gaps, suggesting that semantic retrieval affects natural language flow more than logical structure. This observation points to specific areas for improvement in future RAG systems through better prompt engineering or training objectives that balance faithfulness with natural integration of retrieved content.

3.5.4 Practical Implications

These results provide clear guidance for building surgical QA systems. Semantic retrieval remains necessary despite fluency costs, as the safety implications of unfaithful medical advice outweigh

stylistic concerns. Cross-encoder reranking delivers meaningful improvements that justify the computational overhead. Domain-adapted generators are crucial for faithful context utilization, while fine-tuned retrievers offer only modest benefits over strong biomedical baselines like BioBERT. Our evaluation methodology also yielded important insights. The divergence between traditional NLP metrics (BLEU) and domain-specific evaluation (GPT faithfulness) demonstrates why specialized evaluation frameworks are needed for high-stakes applications. Surface-level similarity metrics may miss critical factual errors that domain-aware evaluation would identify. The key challenge moving forward involves improving how RAG systems integrate retrieved context without sacrificing natural language flow. For medical applications, this requires developing training approaches that maintain factual grounding while producing more naturally flowing text. However, when trade-offs are necessary, factual accuracy must take priority over stylistic considerations in clinical contexts.

3.6 Challenges and Limitations

Our project encountered significant challenges in data accessibility and technical implementation. We initially planned to incorporate the CholecT45 dataset, which provides comprehensive annotations for laparoscopic cholecystectomy procedures, but public accessibility proved problematic with dataset files either unavailable for download or requiring prohibitively large storage capacity of approximately 50GB. This forced us to pivot to collecting surgical videos from YouTube, which provided educational content but lacked systematic annotations. Additionally, we faced a critical technical issue where our fine-tuned generator models produced repetitive, nonsensical output consisting of repeated token sequences such as "ianoianoiano" instead of coherent medical responses. After extensive debugging, we identified the root cause as improper integration of LoRA adapters with the base LLaMA model, where merging and unloading the weights corrupted generation capabilities. The solution involved maintaining adapters as separate layers without merging. Our evaluation was constrained by limited scale and scope. The small corpus size of 1,527 text chunks meant that even random retrieval occasionally performed well, making it dif-

ficult to assess scalability to larger medical document collections necessary for real-world deployment. We also discovered that our "multimodal" approach was actually processing text embeddings from video transcripts using BioBERT rather than visual features, requiring repositioning of our contributions as a text-based system that integrates formal documentation with processed video transcripts. The lack of expert medical validation represents a significant limitation for high-stakes medical applications, and our focus on a single surgical procedure limits generalizability across multiple surgical specialties.

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A Appendix

A.1 Statement of Contributions

This project was collaboratively developed by both authors, who jointly contributed to the conceptualization, design, implementation, and evaluation of the full Retrieval-Augmented Generation (RAG) system for surgical question answering. All major components were developed through close coordination.

- **Retriever Fine-Tuning:** Both authors led the development and training of the dual-encoder retriever, including hard negative mining from BM25 results and domain-specific fine-tuning using BioASQ QA pairs. This involved adapting a BioBERT-based Sentence-Transformer to generate semantically rich biomedical embeddings.
- **Reranker Integration:** Both authors took lead on integrating a cross-encoder reranker (BGE-large) to refine top- k FAISS results. This component significantly improved the contextual relevance of inputs passed to the generator and was evaluated across multiple ablation conditions.
- **Generator Fine-Tuning:** Both authors collaborated on generator fine-tuning using LoRA adapters and FlashAttention to efficiently adapt LLaMA-3.2-11B-Vision-Instruct for text-only QA generation. Prompt engineering, citation grounding, and training loop optimization were conducted jointly.
- **Evaluation Dataset Construction:** Together, we created a novel evaluation dataset of 32 laparoscopic cholecystectomy QA pairs using GPT-4. We co-developed the prompting strategy, manually validated answers against trusted surgical sources, and annotated each question with semantic tags for downstream analysis.

- **Evaluation Framework:** Both authors implemented the full evaluation pipeline, including traditional metrics (BLEU, ROUGE-L, BERTScore), factuality checks (FactCC, MNLI entailment), and a custom LLM-as-a-Judge loop using GPT-4o-mini to score context precision, faithfulness, fluency, and coherence.

- **Ablation Studies and Analysis:** Both authors collaborated on designing the ablation matrix and interpreting model behavior across retrieval, reranking, and generation configurations. We analyzed key trade-offs between fluency and factuality, grounding our insights in both metric scores and qualitative review.

- **Writing and Reporting:** The final paper, appendices, and documentation were written collaboratively. We co-authored all sections, jointly edited the text, and ensured consistency between implementation and claims. Visualizations and tables were created through shared effort.

This project represents an equal and collaborative effort, with both authors deeply involved in all technical and research decisions throughout the development lifecycle.

A.2 Retriever Fine-Tuning

To construct a domain-adapted retriever for biomedical and procedural question answering, we fine-tuned a dense embedding model using supervised contrastive learning on the BioASQ 13b dataset. We selected pritamdeka/BioBERT-mnli-scnli-scnli-scitail-mednli-stsb as the base encoder for its extensive pretraining on PubMed and biomedical NLI datasets, ensuring strong domain awareness. The goal was to learn a shared embedding space in which semantically related questions and context passages are mapped closely together, enabling effective semantic retrieval using inner product similarity.

We first extracted QA pairs from BioASQ JSON files. Each question was paired with its corresponding “snippets” — biomedical context passages curated by domain experts. Snippets were filtered to remove duplicates and trimmed for length. We retained only those context passages that were non-trivial and semantically dense. To

improve the model’s ability to distinguish semantically adjacent but irrelevant content, we implemented hard negative mining using BM25. Specifically, for each question, we used the rankbm25 algorithm to retrieve the top 50 context passages from a pool of all snippets, excluding the true positive. From these, we randomly sampled 5 hard negatives per question, producing a challenging contrastive setup.

The training process used the SentenceTransformerTrainer module from the sentence-transformers library. Each batch was constructed using MultipleNegativesRankingLoss, where each question had one positive and multiple hard negative context passages. The model was trained using a batch size of 64, learning rate of 2e-5, and warmup ratio of 0.1 over 5 epochs. The training set was shuffled at each epoch, and reproducibility was ensured through fixed seeds across PyTorch, NumPy, and CUDA. Upon training completion, the learned embedding model was evaluated via FAISS (IndexFlatIP) to ensure efficient semantic retrieval using inner product scoring. This retriever served as the front-end to our SurgiRAG pipeline, enabling high-recall retrieval over procedural and biomedical contexts at inference time.

A.3 Generator Fine-Tuning

The generator in our RAG system is based on the meta-llama/Llama-3.2-11B-Vision-Instruct model, which we fine-tuned in a text-only configuration to improve factual accuracy and clinical relevance in generated answers. While the base model supports multimodal input, we deliberately constrained it to process only textual context—namely, surgical video transcripts and structured procedural documentation. The fine-tuning objective was to enhance the model’s ability to generate grounded, coherent answers in response to biomedical questions, conditioned on retrieved context passages. To do this efficiently, we used Low-Rank Adaptation (LoRA), enabling us to fine-tune the model using limited GPU resources without updating the full parameter set.

Training examples were derived from the BioASQ 13b dataset, which contains expert-annotated biomedical question-answer pairs with supporting context. We implemented a custom data processor that filtered out yes/no questions, very short answers, or examples with missing snippets. For each eligible example, we con-

structed inputs that simulate real-world RAG behavior. The final prompt included a system message and user query, along with up to five retrieved passages formatted as [Passage i]: When available, irrelevant snippets were added to train the model to recognize and abstain when presented with insufficient information. In such cases, the expected output was a special abstention message: ”The passages do not contain sufficient information to answer.”

The tokenizer handled prompt construction using the HuggingFace AutoTokenizer in chat-template mode, simulating a multi-turn assistant interaction. Token sequences were truncated to a maximum length of 768 tokens, with care taken to preserve the full target answer and as much of the prompt as possible. Tokenized sequences were padded using a custom PyTorch collator that masked labels for the prompt portion, ensuring that the model learned only from the response portion of the sequence.

The fine-tuning was conducted using the HuggingFace Trainer API, with mixed-precision training (bfloat16) enabled for memory efficiency. We used a batch size of 1 (per device), gradient accumulation of 2 steps, a cosine learning rate scheduler with warmup ratio of 0.1, and a maximum of 3 training epochs. To improve generalization and prevent overfitting, early stopping with a patience of 5 evaluations was applied, and evaluation checkpoints were saved every 200 steps. Importantly, LoRA adapters were applied selectively to attention projection layers (q_proj, k_proj, v_proj, o_proj), and we avoided merging them into the base model weights post-training due to observed decoding instability when doing so.

Evaluation of the fine-tuned model involved both traditional metrics—BLEU, ROUGE-L, and BERTScore—as well as entailment-based factuality scoring using a DeBERTa-based MNLI classifier. The evaluation pipeline also computed metrics such as average prediction length and reference-to-output length ratios. This generator, when paired with our domain-adapted retriever and reranker, forms the core of the SurgiRAG system, producing grounded, concise, and clinically appropriate responses to surgical questions.

A.4 Prompt Engineering Strategy

To guide the generator toward producing grounded and clinically appropriate responses, we de-

signed a custom instruction-style prompt format named `SYSTEM_PROMPT_FEWSHOT`. This prompt plays a critical role in shaping the behavior of the language model during inference, particularly when integrating retrieved evidence in a constrained medical QA setting.

The prompt begins with a high-level instruction directing the model to “answer the query based on the following context items” and explicitly encourages the model to internally process the input by extracting relevant passages before formulating a response. Crucially, it also includes a constraint: the model must not reveal its reasoning process, but instead produce a fluent, explanatory answer as if written by a medical expert. This mitigates verbosity and hallucination risks that arise from chain-of-thought-style generations in sensitive domains.

To anchor the model’s output format and tone, we included two few-shot exemplars demonstrating domain-specific answers. These examples are carefully chosen to reflect both factual specificity and fluent explanation. The first addresses a high-risk surgical complication (“bile duct injury”) and models a clinically precise yet readable answer. The second covers a procedural concept (“critical view of safety”) and emphasizes anatomical detail while maintaining concise phrasing. These examples help condition the model to strike a balance between factual grounding and natural clinical language.

Following the examples, the final instruction reiterates the structure expected in the generation phase: the model is told to use the provided “Context” items to generate the answer, and to cite supporting evidence inline using numeric references such as [1], [2], etc. This citation mechanism reinforces attribution and encourages the model to remain faithful to retrieved content rather than rely solely on internal knowledge.

The prompt serves multiple purposes simultaneously: it primes the model to answer like a surgical educator, promotes factual grounding through retrieval alignment, discourages hallucinated reasoning paths, and sets a clear stylistic precedent. By embedding few-shot exemplars that mirror the evaluation domain, we reduce distributional mismatch between training data (BioASQ) and target inference queries (surgical procedural QA). In practice, this prompt was used during ablation testing across all generation configurations to en-

sure consistent answer structure and stylistic fidelity.

A.5 LLM-as-a-Judge Evaluation Methodology

To complement traditional NLP metrics and more effectively assess factual grounding in surgical question answering, we implemented an asynchronous **LLM-as-a-Judge** framework using `gpt-4o-mini`. This method allows automatic scoring of qualitative dimensions—*faithfulness*, *context alignment*, *fluency*, and *coherence*—across 32 samples per model variant under strict API rate limits.

System Setup. Evaluation calls were made using an `AsyncOpenAI` client, wrapped in an `AsyncLimiter` to throttle to 60 requests per minute. All evaluations were run with fixed seeds for reproducibility. Each GPT call used strict prompting under a system role (“You are a strict evaluator. Answer **ONLY** with the requested output.”) to enforce output constraints and reduce LLM hallucinations. A retry mechanism was implemented to handle latency and timeout errors.

Evaluation Metrics. Each axis was evaluated using a specialized prompt template:

- **Context Precision:** Measures whether retrieved chunks are collectively relevant to the question. GPT is shown the question and up to 5 context chunks (truncated to 300 characters each) and asked:

```
``Do these chunks
collectively help answer
the question?  Reply only
YES or NO.``
```

A “YES” maps to 1.0, “NO” to 0.0.

- **Faithfulness Score:** Evaluates whether the generated answer is fully supported by the provided context. GPT is shown both and asked:

```
``Does the context fully
support every factual claim
in the answer?  Respond with
a single number 0--1.``
```

The response is parsed as a floating-point number.

- **Fluency and Coherence:** Rates stylistic quality on a 1–5 scale. The prompt is:

```

Rate the answer below on a
1--5 integer scale for each
trait.
Traits: (A)
Fluency/Grammar, (B)
Coherence/Logical-flow.
Return JSON exactly
like {'fluency':4,
'coherence':5}.

```

Results are parsed into two numerical dimensions.

Benefits and Limitations. This methodology allowed us to uncover qualitative patterns not captured by overlap-based metrics like BLEU or BERTScore. For instance, we observed that configurations using random retrieval scored highly on fluency (3.75) and coherence (4.16), despite extremely low faithfulness and Precision scores (0.25 and 0 respectively), highlighting how language models may fall back on pretrained priors when no relevant evidence is provided.

However, this framework is not without limitations. Although gpt-4o-mini performs reliably in this evaluation setup, it is still an automated model-based proxy for human judgment. While consistent across experimental runs, it lacks clinical expertise and may miss subtle domain errors. As such, future work should incorporate human evaluations by surgical professionals as a gold standard, particularly for assessing safety-critical outputs.

A.6 Evaluation Dataset Construction via GPT-4

To evaluate our surgical Retrieval-Augmented Generation (RAG) system in a controlled and domain-relevant manner, we constructed a high-quality evaluation set of 32 laparoscopic cholecystectomy (LC) question-answer pairs using GPT-4. This decision was driven by the lack of publicly available QA datasets tailored to surgical procedural knowledge, as most existing benchmarks (e.g., BioASQ, MedQA, PubMedQA) are not fine-grained or clinically specific enough for laparoscopic surgery.

Dataset Composition. The dataset was generated iteratively in conversation with ChatGPT-4 using a carefully crafted system prompt to emulate the style and factual grounding of expert-written surgical guides. Each QA pair consists of:

- A precise clinical or procedural **question** relevant to LC.

- A concise, explanatory **answer** that emulates the tone of surgical board review materials.
- An inline **citation**, typically to PubMed-indexed articles (e.g., PMID:28846328), clinical guidelines (e.g., Tokyo 2018), or formal surgical societies.
- A **semantic category** tag such as Complications, Contraindications, or Technique for future stratified evaluation.

Prompting and Curation. The QA generation was guided by a prompt that emphasized educational clarity, clinical correctness, and citation grounding. All GPT-generated responses were manually verified for:

- **Factual accuracy** against primary literature and surgical manuals.
- **Clarity and conciseness** suitable for expert-level QA benchmarks.
- **Relevance** to intraoperative decision-making and surgical best practices.

Utility. This dataset served as our gold standard for evaluating generative answer quality across multiple configurations of our RAG system. It was used in:

- **Few-shot prompting** during generator fine-tuning.
- **Ablation studies** of retrieval and reranking components.
- **Automated evaluation** using both traditional NLP metrics (BLEU, ROUGE, BERTScore) and LLM-as-a-judge metrics (faithfulness, fluency, coherence).

We instructed GPT-4 to answer questions in a concise, explanatory style appropriate for surgical education. All answers were cross-referenced against trusted sources (e.g., NIH guidelines, surgical textbooks) and manually curated to ensure clinical correctness and citation fidelity. Questions were diverse in scope and difficulty, targeting both foundational knowledge (e.g., "What is the critical view of safety?") and nuanced scenarios (e.g., "Is laparoscopic cholecystectomy safe in compensated cirrhosis?").

This dataset served as the gold standard for evaluating the factuality, fluency, coherence, and groundedness of our generated outputs. It also enabled controlled few-shot prompting, reproducible ablation studies, and multi-metric benchmarking using both traditional and LLM-as-a-judge scoring methods.

By creating our own evaluation set tailored to the laparoscopic cholecystectomy domain, we ensured domain-relevance, clinical specificity, and the ability to probe fine-grained distinctions in generative fidelity that generic biomedical QA datasets could not capture.

Reproducibility Checklist

We include the following reproducibility information to support the replication and extension of our results.

- **Code Availability:**

All code for data preprocessing, model training, retrieval indexing, and evaluation is available at: [<https://github.com/Vinithkuruppu/266-final-project>].

- **Data Sources:**

- **Training Data:** BioASQ 13b question-answer pairs with supporting snippets.
- **Evaluation Corpus:** 17 held-out surgical documents and transcripts (409 pages, 1,527 chunks).
- **Custom QA Benchmark:** 32 surgically-relevant questions covering procedures, anatomy, and safety practices.

- **Hardware Used:**

- Generator fine-tuning was performed on a single NVIDIA A100 40GB (Google Colab).
- Retriever fine-tuning was run on a single NVIDIA A100 40GB (Google Colab).
- Evaluation using GPT-4o-mini ran asynchronously using OpenAI API with a rate limit of 60 requests/minute.

- **Model Architectures:**

- **Retriever:** SentenceTransformer initialized from `pritamdeka/BioBERT-mnli-snli-scinli-scitail-mednli-stsb`.

- **Reranker:** Cross-encoder from `BAAI/bge-reranker-large`.

- **Generator:**

`meta-llama/Llama-3.2-11B-Vision-Instruct`, used in text-only mode with LoRA fine-tuning.

- **Fine-Tuning Details:**

- **Retriever:** 5 epochs, batch size 64, learning rate $2e-5$, hard negatives from BM25 (top-50 pool).
- **Generator:** 3 epochs, learning rate $5e-5$, LoRA ($r = 16$, $\alpha = 32$), max sequence length 768, gradient accumulation = 2.
- **Optimization:** AdamW (paged 8-bit for generator), cosine LR scheduler, warmup ratio 0.1.
- **Precision:** Generator trained with `bfloat16` and FlashAttention enabled.

- **Evaluation Setup:**

- **Overlap Metrics:** BLEU, ROUGE-L, BERTScore.
- **Faithfulness Metrics:** FactCC, MNLI-based entailment, GPT-4o-mini scoring.
- **GPT LLM-as-a-Judge:** Evaluated context precision, faithfulness (0–1), fluency (1–5), and coherence (1–5) with strict prompts.
- **Evaluation Sample Size:** All 12 pipeline configurations were evaluated on the same 32-question benchmark.

- **Random Seeds:** `random.seed(42)`, `numpy.seed(42)`, `torch.manual_seed(42)` were used in all training scripts.

- **Trained Weights:** LoRA adapter weights and retriever checkpoints can be shared upon request.

- **Prompt Templates:**

- **Generator Prompt:** Instructional format with few-shot QA examples, citation requests (e.g., “[1]”), and system roles.
- **LLM-as-a-Judge Prompts:** Strictly structured prompts returning YES/NO, 0–1 scores, or JSON outputs.